Fields and Waves I

Lecture 9

Fields & Math II

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Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Divergence
- Curl
- Gradient
- Electrostatics



Field Math

- As mentioned in the previous lecture, **we have now entered the "fields" portion of the class.**
- We discussed three coordinate systems that we will use often this semester: cartesian (rectangular), cylindrical and spherical
- Now we must discuss the vector operators: divergence, curl and gradient
- *Note*: the form of each differs by coordinate system!

$\nabla . E = \frac{\rho}{\epsilon_0}$	(1)	Gauss' law
$\nabla . B = 0$	(2)	Magnetic monopoles
$\nabla \times E = -\frac{\partial B}{\partial t}$	(3)	Faraday's law
$\nabla \times H = J + \frac{\partial D}{\partial t}$	(4)	Ampere-Maxwell law

Divergence

Field Math

Gauss' Law: Electric charge gives rise to an electric field.

Differential Form:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$\vec{D} = \varepsilon \vec{E}$

$\nabla \cdot \vec{D} = \rho$

divergence

"rho" charge density

epsilon

Integral Form:

$$\oiint_S \mathbf{E} \cdot \mathrm{d}\mathbf{A} = \frac{\mathrm{Q}}{\epsilon_0}$$



Source: spectrumscientifics.wordpress.com

Field Math

What does the field an electric point charge gives rise to actually look like?

https://davidawehr.com/projects/electric_field.html

Field Math

- Points of electric charge are **sources** or **sinks** of electric field lines.
- In mathematical terms, we say that the **divergence** of an electric field at a point is proportional to how much charge it has.

"del" or "nabla"

Notation: $div(\vec{A}) = \nabla \cdot \vec{A}$ *NOT a DOT product but has similar features*

Result is a **SCALAR**, composed of derivatives

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$ in Cartesian coordinates

$\vec{\nabla} = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle$

Field Math

Divergence Theorem:

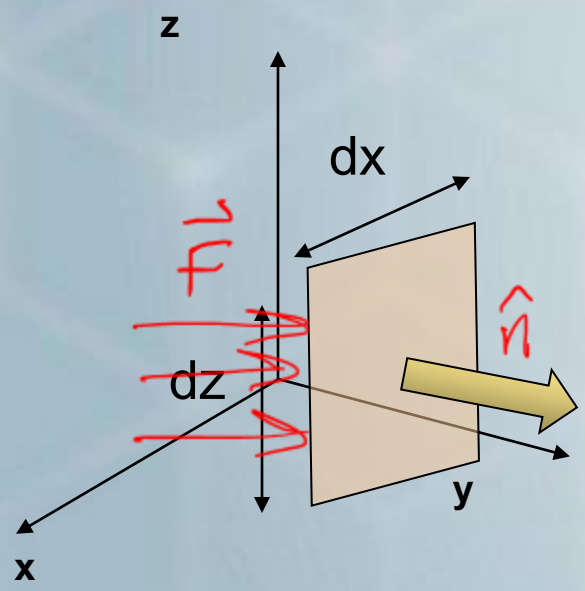
$$\oint \vec{A} \cdot \vec{ds} = \int (\nabla \cdot \vec{A}) \cdot dV$$

Volume integral on right is volume enclosed by surface on the left

In plain English:

"The amount and magnitude of field sources or sinks inside a volume will determine the net flux through the volume's surface."

Field Math



- An important property in field calculations is **flux**: how *much* field is flowing through a surface.
- Whenever we have a vector field, we can calculate a flux by taking the following integral:

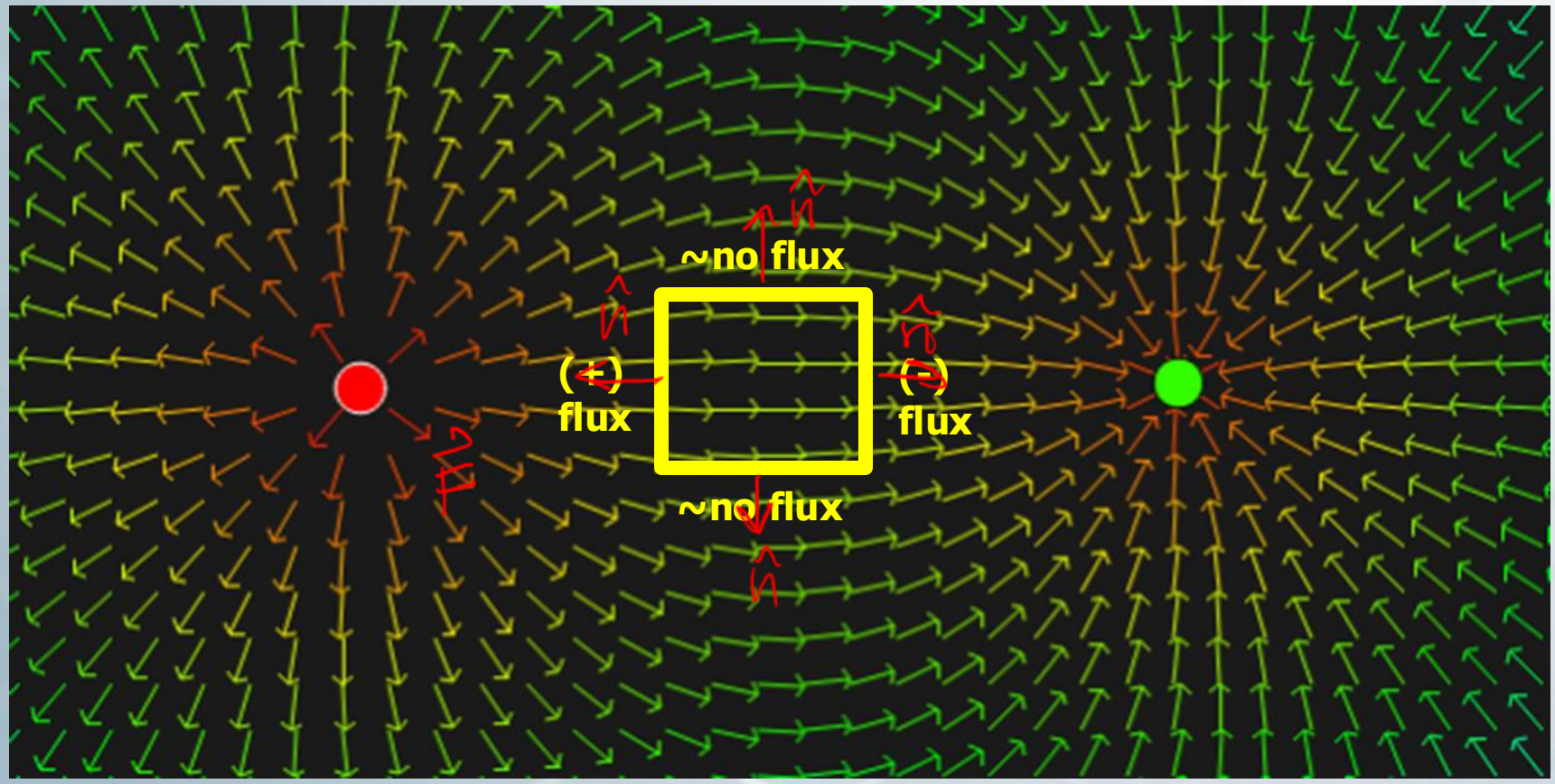
$$\text{Flux} = \iint_S \vec{\mathbf{F}} \cdot \hat{\mathbf{n}}\, dS$$

surface area element

- Note that the total flux will depend on the **area of the surface** through which the field is flowing, the **magnitude of the field**, and the **angle between the field vectors and the surface normal**.

Field Math

Surface integrals and flux

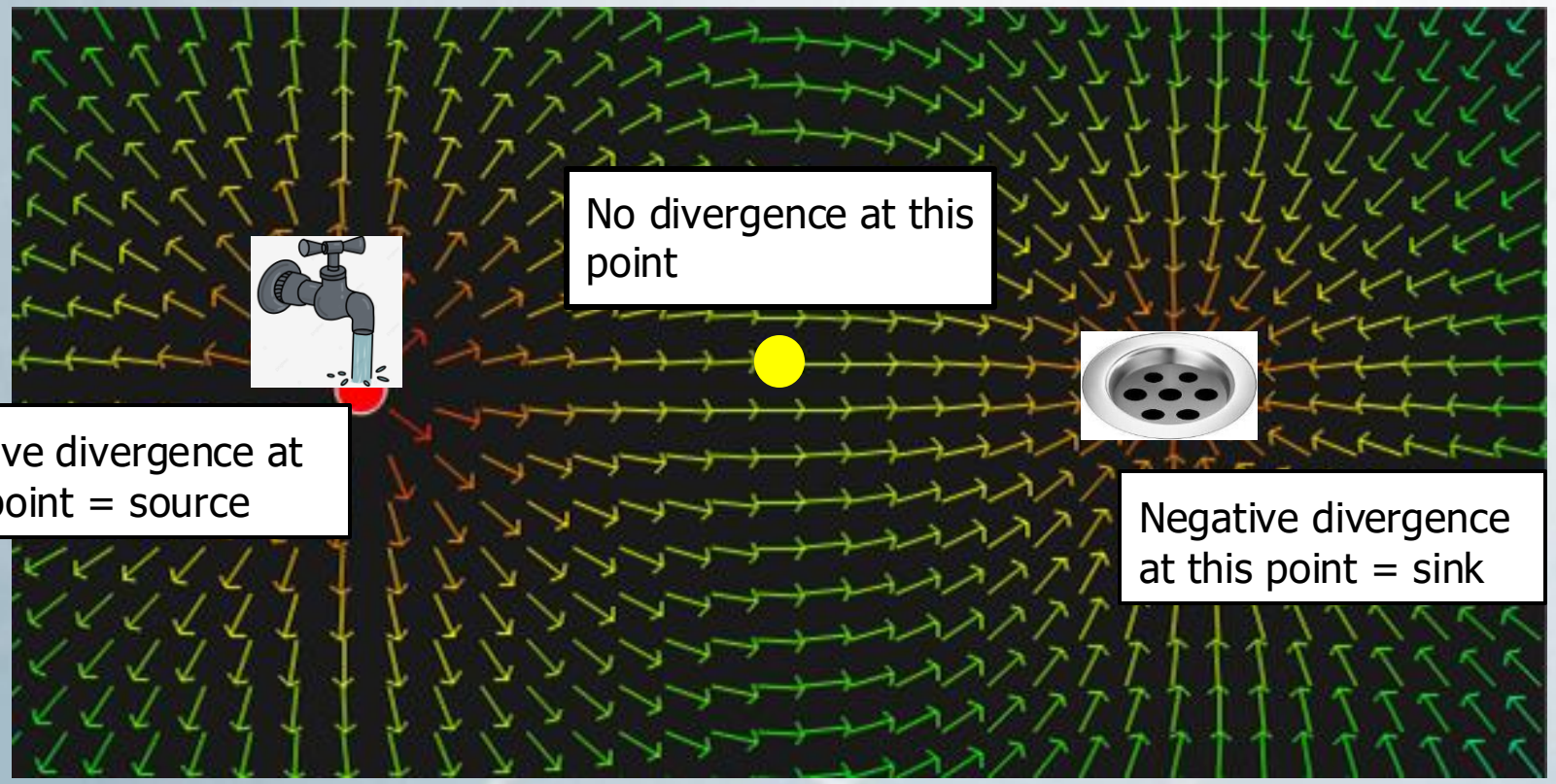


What is the net flux through the box in the image above?

Zero. But how do we know?

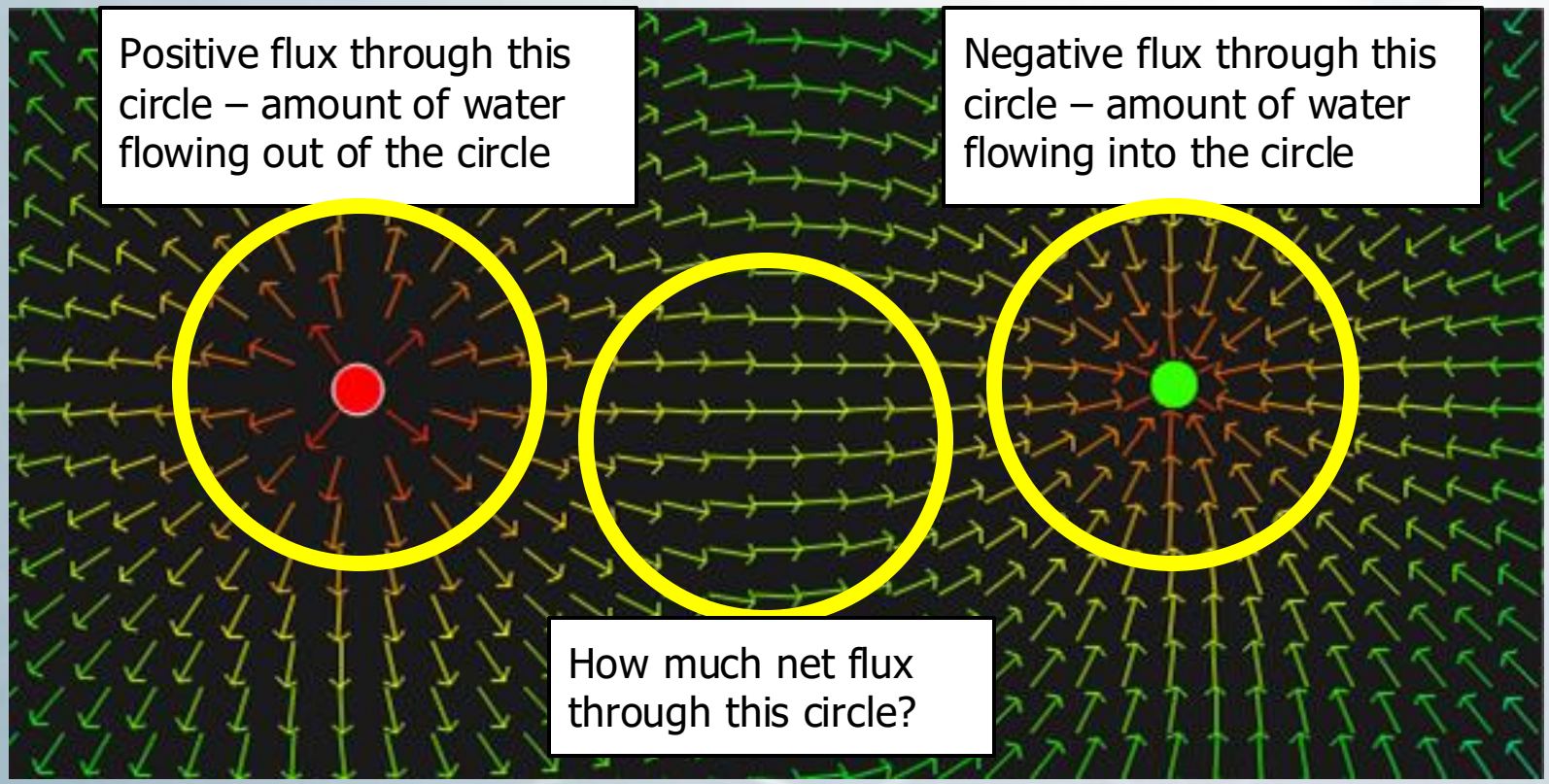
Field Math

Imagine that this plot is a vector field of water flow in 2D

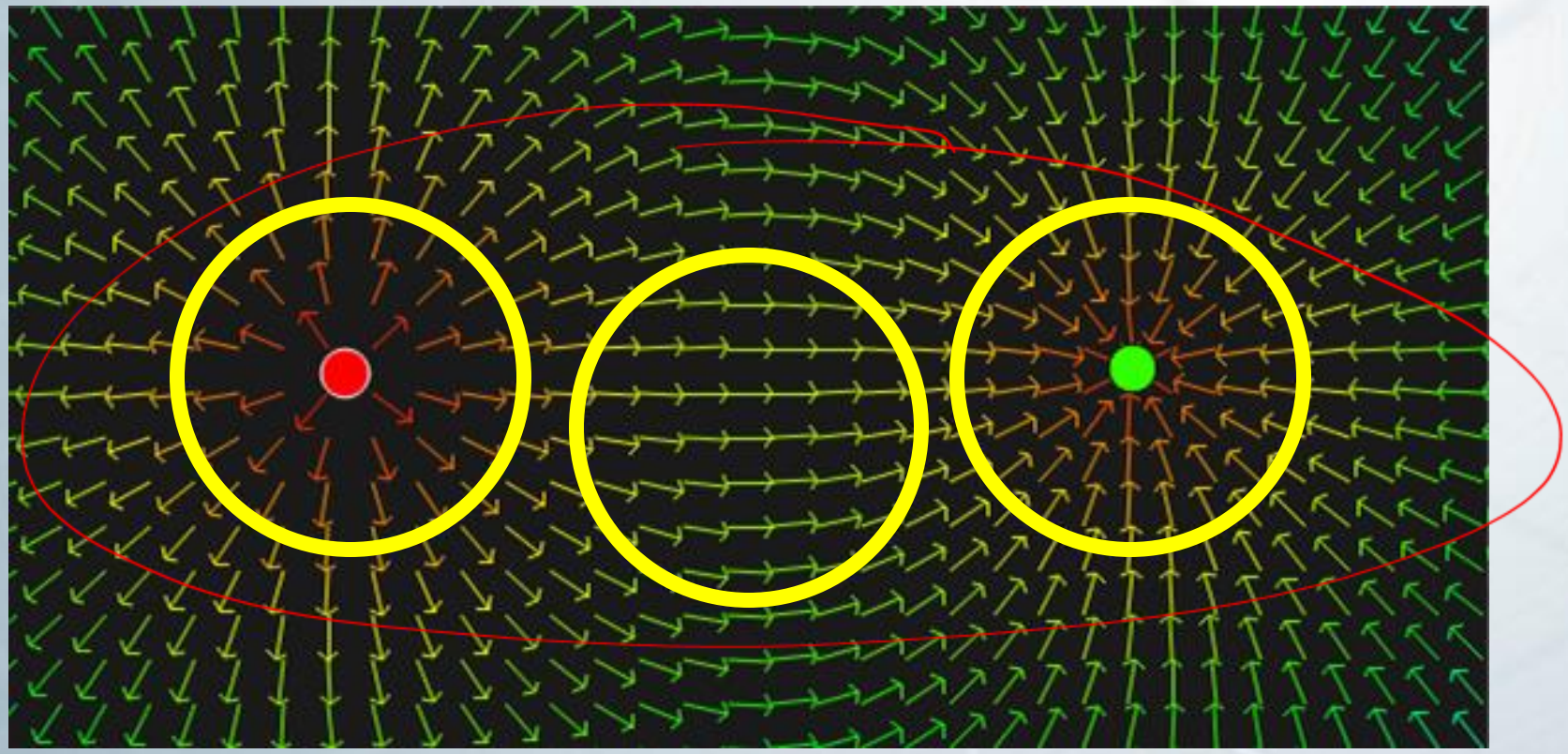


Field Math

Flux = magnitude of flow through a surface



Field Math



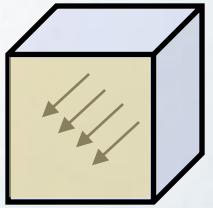
- You can see that the flux through any of these three circles has a relationship to the amount of "charge" (source or sink of vector field) inside.
- Conservation laws stipulate that the **amount of water entering or leaving** through the boundaries of the the circle per unit time has to be **equal to the total change in the amount of water inside the circle per unit time**.

Field Math

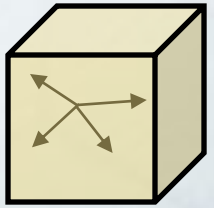
Divergence = degree to which a given point is a source or a sink

"Global" quantities

$\int \vec{A} \cdot d\vec{S}$ Measures flux through any surface



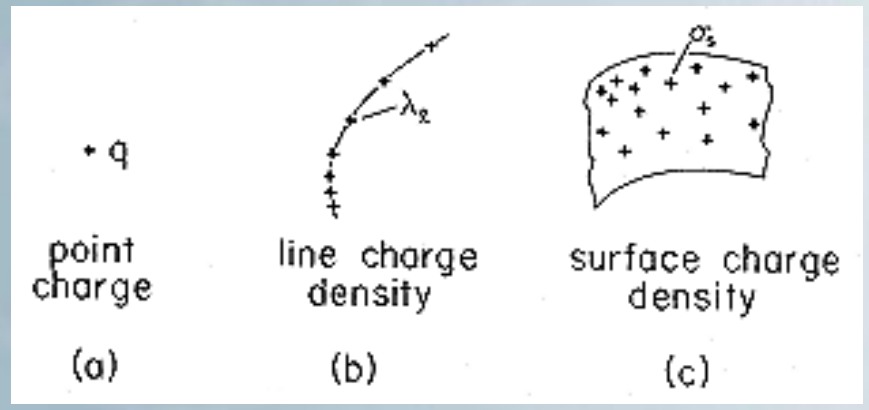
$\oint \vec{A} \cdot d\vec{S}$ Measures flux through **closed** surfaces



is related to via Divergence theorem

$\nabla \cdot \vec{A}$ is a "local" measure of flux through an infinitesimal volume

Field Math



https://web.mit.edu/6.013_book/www/chapter1/1.3.html

$$\vec{A} = \varepsilon \vec{E}$$

$$\oint \vec{A} \cdot d\vec{s} = \int (\nabla \cdot \vec{A}) \cdot dV$$

Q

ρ

One last important point:

The source of flux could be a ***single point*** **with a nonzero divergence**, or the divergence could come from a **line**, **surface** or **volume**. This parallels the fact that electric charges can be treated as points, but it is often more accurate to speak of a ***region of charge distribution***.

Field Math

Divergence operator simulation:

https://em8e.eecs.umich.edu/jsmodules/ch3/mod3_3.html

Plot the divergence of the field: $\boldsymbol{A} = \boldsymbol{x}^2\hat{\boldsymbol{x}} + \boldsymbol{y}^2\hat{\boldsymbol{y}}$

- Is the divergence proportional to where the change in the field is largest?
- Is the divergence zero anywhere? Does this make sense in terms of the "local" flux interpretation of divergence?

Curl

Field Math

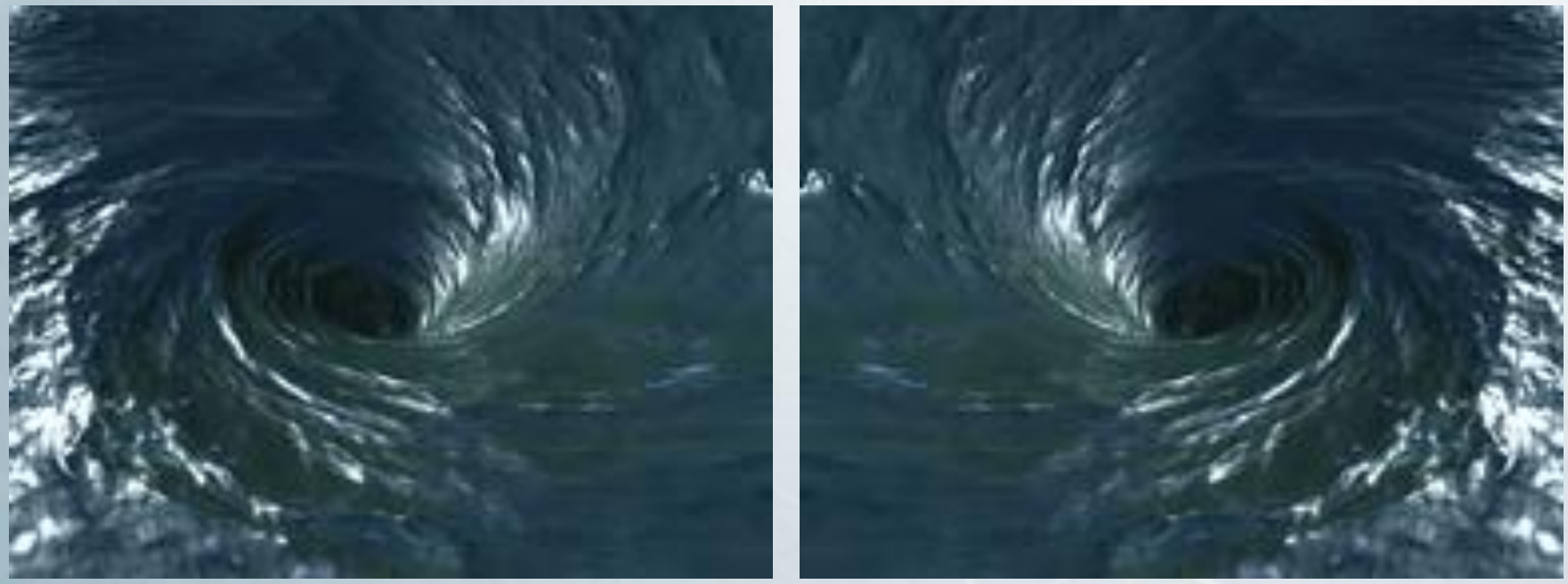
What's the difference between these two whirlpools?





Field Math

What's the difference between these two whirlpools?



Circulation in opposite directions.
Their velocity fields have opposing curl.

Field Math

- Curl can be calculated at a point using the following expression for Cartesian coordinates (Ulaby pg. 158).

- Similar expressions exist for the other coordinate systems. Curl direction is given by right-hand rule.

$$\nabla \times \mathbf{B} = \hat{\mathbf{x}}\left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z}\right) + \hat{\mathbf{y}}\left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x}\right) + \hat{\mathbf{z}}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)$$

$$= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ B_x & B_y & B_z \end{vmatrix}. \qquad (3.105)$$

Field Math

- The curl operator

NOTATION: $\nabla \times B$ Result of this operation is a VECTOR

This is NOT a CROSS-PRODUCT

- Stokes's theorem:

$$\oint \vec{B} \cdot \vec{dl} = \int\int (\nabla \times \vec{B}) \cdot ds$$

line integral curl surface integrated over ds

Surface integral on right is surface enclosed by line on the left

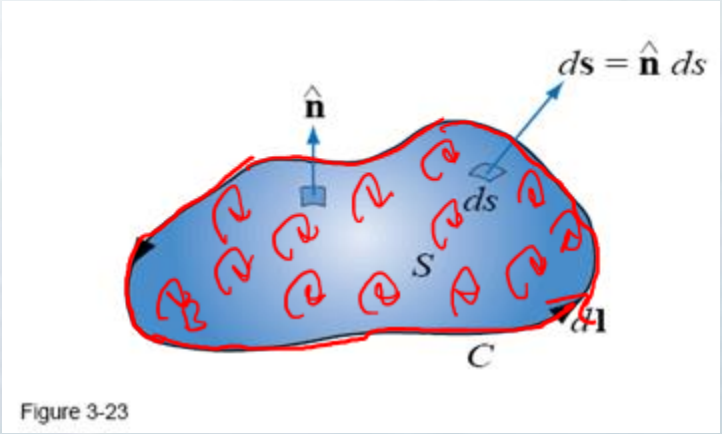


Figure 3-23

Field Math

$$\oint \vec{B} \cdot \vec{dl} = \int \int \nabla \times \vec{B} \cdot ds$$

In plain English:

"The amount of B pointing in a loop around the outside of surface ds is determined by the total curl of field B inside ds."

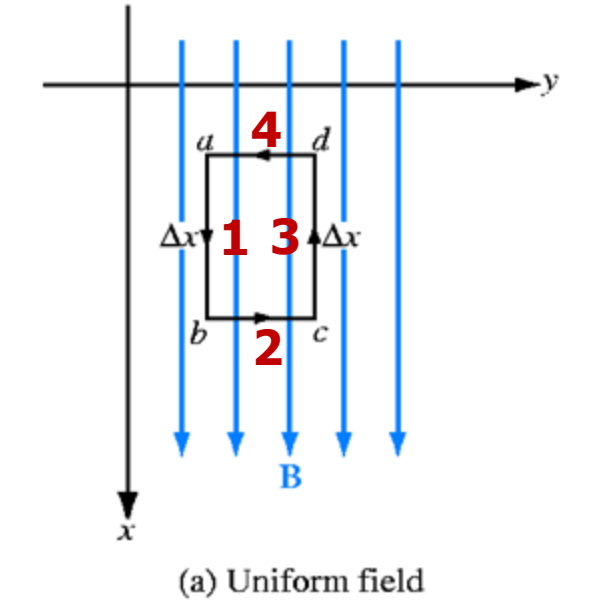
OR

"If you drop a ball near a whirlpool, the stronger the whirlpool is, the faster the ball will travel around it."



Field Math

NOTATION : $\oint$ Implies a CLOSED LOOP Integral



(a) Uniform field

$$\oint \vec{B} \cdot \vec{dl}$$ measures circulation (related to Curl)

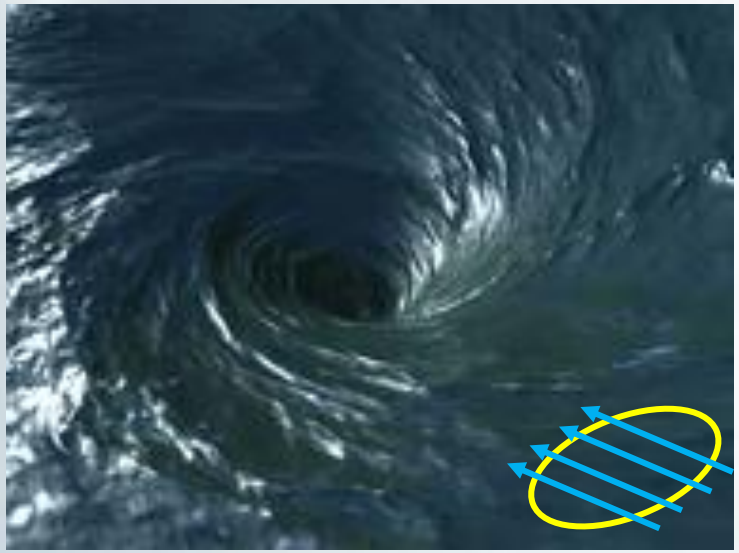
Example of a uniform field B in the x direction

$$\text{circulation} = \oint \vec{B} \cdot \vec{dl} =$$

$$\int_a^b B\hat{x} \bullet \hat{x}dx + \int_b^c B\hat{x} \bullet \hat{y}dy + \int_c^d B\hat{x} \bullet \hat{x}dx + \int_d^a B\hat{x} \bullet \hat{y}dy = 0$$

1 2 3 4

Field Math



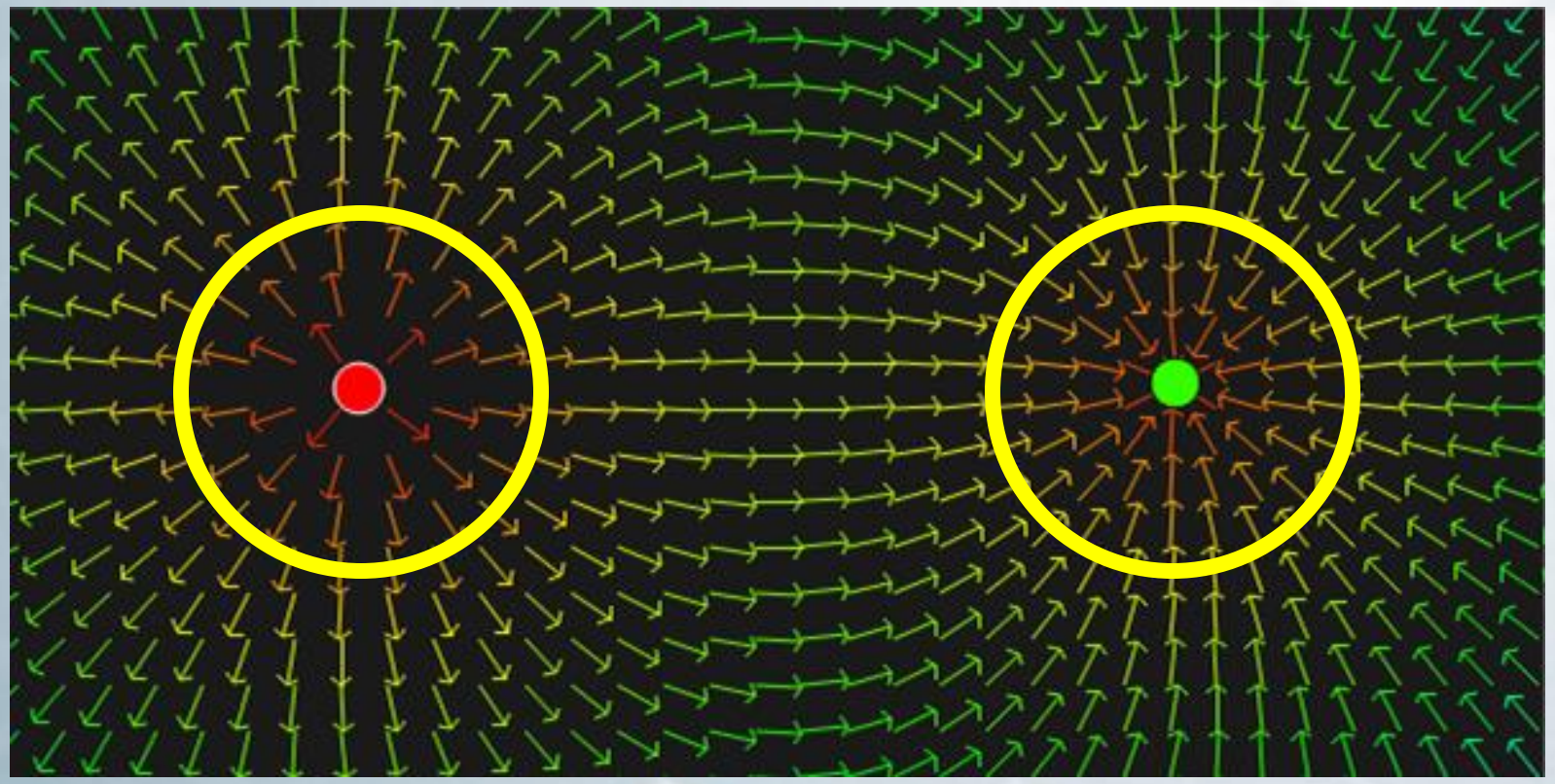
- The eye of the whirlpool is a place where the curl is nonzero.

  Whereas the point charges were *sources* or *sinks* of field lines, the whirlpool eye is a point that causes ***circulation*** of a vector field (velocity) around it.

zero

- What is the curl inside the yellow circle? If you drop a "paddlewheel" into the center of the yellow circle, will it rotate?

Field Math



What is the curl of the field inside these two circles?

zero

Field Math

One last note on whirlpools:



- We previously mentioned that flux is often due to a *distribution* of divergence.
- Circulation of a vector field in practice is often due to a ***distribution* of curl** - that is, the vector field acts less like it has one whirlpool and more like it has a **distribution of infinitesimally-small whirlpools**.

Field Math

Curl operator simulation:

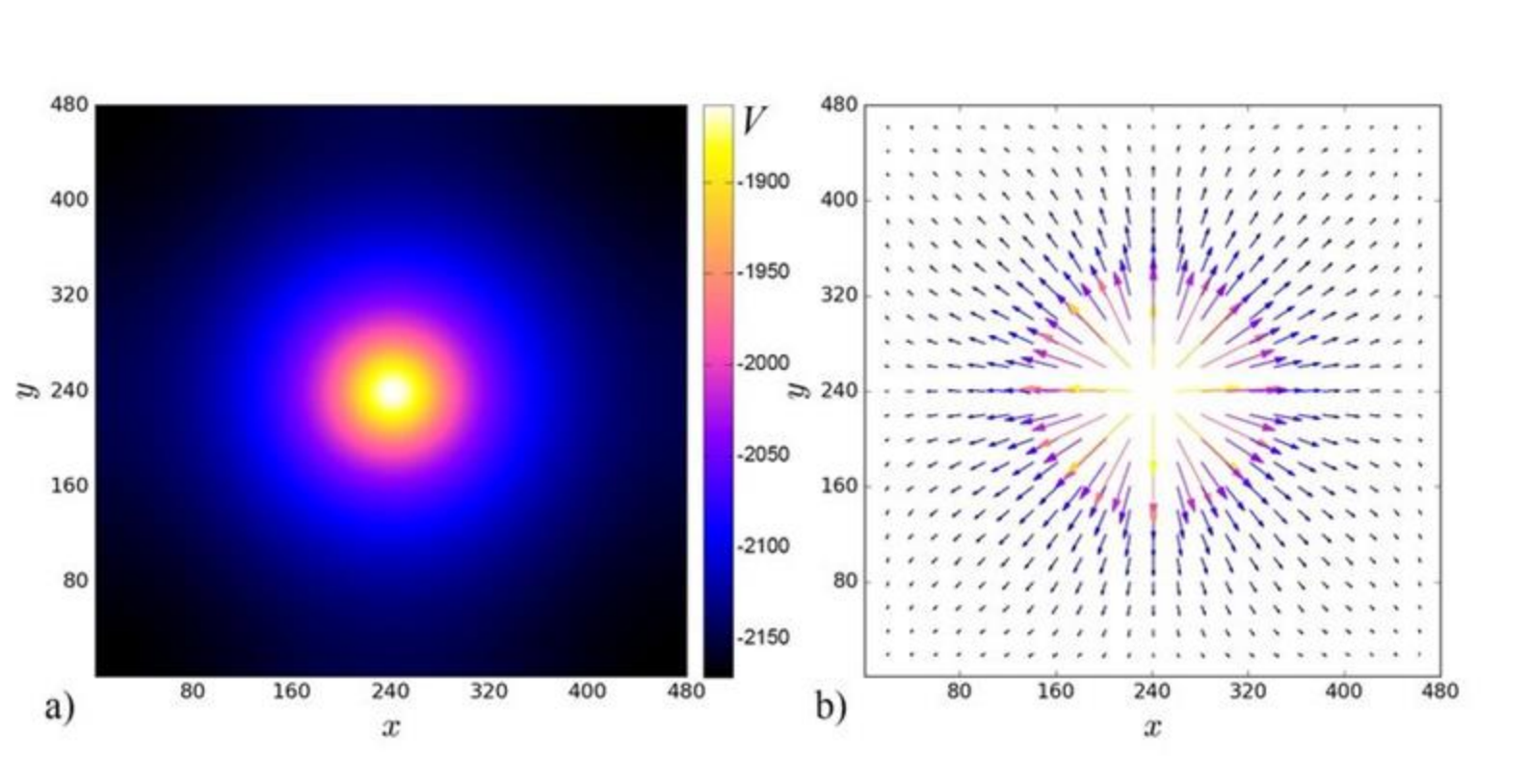
https://em8e.eecs.umich.edu/jsmodules/ch3/mod3_4.html

Plot the curl of the field: $\boldsymbol{A} = \boldsymbol{y}^2\hat{\boldsymbol{x}} + \boldsymbol{x}^2\hat{\boldsymbol{y}}$

- Is the curl higher in regions in which the field lines seem to rotate more?
- Is the curl zero anywhere? Does this make sense in terms of the "local" rotation interpretation of curl?

Gradient

Field Math



What is the relationship between the potential field (left) and the electric field (right)?

The electric field is defined by the gradient of the potential field.

Field Math

- The gradient operator

  GRADIENT measures **CHANGE** in a **SCALAR FIELD**

  - The result is a VECTOR pointing in the direction of greatest increase
  - For a Cartesian system:

$$\nabla \mathrm{f} = \frac{\partial f}{\partial x}\hat{x} + \frac{\partial f}{\partial y}\hat{y} + \frac{\partial f}{\partial z}\hat{z}$$

- Main property

You will find that $\nabla \times (\nabla f) = 0$ ALWAYS

$\nabla\phi = -\vec{E}$

Field Math

Gradient operator simulation:

https://em8e.eecs.umich.edu/jsmodules/ch3/mod3_2.html

Plot the gradient of the field: $\mathbf{A} = \mathbf{x}^2 + \mathbf{y}^2$

- Is the magnitude of the gradient proportional to the change in the length of the vectors?
- Is the gradient zero anywhere? Does this make sense in terms of the "local" rate of change of the field magnitude?

Field Math

- **Divergence**
  - Measures the flux of a vector field through a surface

$$divA \text{ or } \nabla \bullet A$$ Result is a SCALAR

- **Curl**
  - Measures the circulation of a vector field

$$curlB \text{ or } \nabla \times B$$ Result is a VECTOR

- **Gradient**
  - Measures the change in a scalar field

$$grad(f) \text{ or } \nabla f$$ Result is a VECTOR

Do Lecture 9 Exercise in groups of up to 4.

Electrostatics

Electrostatics

We will now begin applying the field math to solving Maxwell's equations. But given the complexity of this task, we will start with a simplified case: **electrostatics**.

- Charges do not move
- No current
- As a consequence, there is no magnetic field; only an electric field

Electrostatics

Full Version of Maxwell's Equations

Integral Form	Differential Form
$\oint \vec{B} \cdot d\vec{S} = 0$	$\nabla \cdot \vec{B} = 0$
$\oint \vec{D} \cdot dS = \oint \rho dv = Q_{encl}$	$\nabla \cdot \vec{D} = \rho$
$\oint \vec{H} \cdot d\vec{l} = \int \vec{J} \cdot d\vec{S} + \int \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$
$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{S}$	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Electrostatics

Electrostatic Version of Maxwell's Equations

Integral Form	Differential Form
$\oint \vec{D} \cdot d\vec{S} = \int \rho dV$	$\nabla \cdot \vec{D} = \rho$
$\oint \vec{E} \cdot d\vec{l} = 0$	$\nabla \times \vec{E} = 0$

These equations represent Gauss's Law (just written in a different form)

Electrostatics

Integral Form	Differential Form
$\oint \vec{D} \cdot d\vec{S} = \int \rho dV$	$\nabla \cdot \vec{D} = \rho$
$\oint \vec{E} \cdot d\vec{l} = 0$	$\nabla \times \vec{E} = 0$

These equations tell you that in electrostatics, the E-field has no curl (and there are no net voltages around loops).

Electrostatics

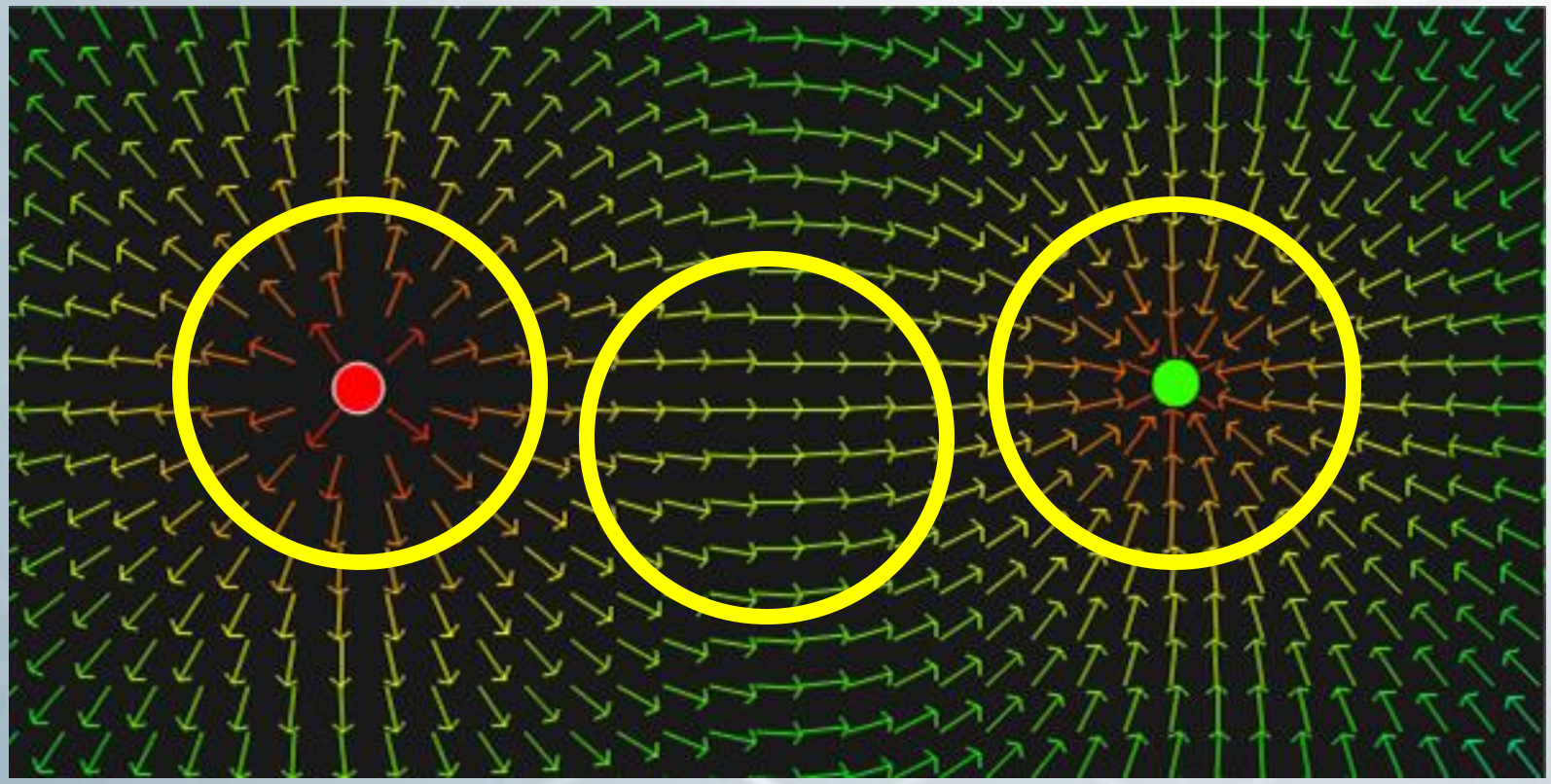
There are 2 different ways of referring to the electric field.

$\vec{E}$ (often called just called "electric field" or "E-field") has units of volts per meter (V/m). Thus it has a direct relationship with voltage.
$\vec{D}$ (often called "electric displacement field"), has units of coulombs per square meter (C/m^2) and thus has a direct relationship with charge or charge density.

The two quantities are related by: $\vec{D} = \epsilon \vec{E}$

$\boldsymbol{\varepsilon}$ represents *permittivity*, a property of the medium that the electric field is present in/propagating through.

Electrostatics



Recall these three circles containing different amounts of charge.

Electrostatics

$$\nabla \cdot \vec{D} = \rho$$

We had stated that the points that contain charge have a non-zero divergence, whereas the points with no charge have a divergence of zero.

This is the differential form of Gauss's Law.

Electrostatics

$$\oint \vec{D} \cdot d\vec{S} = \int \rho dV$$

This could take the form of point charges or line, surface, or volume charge density

The integral form of Gauss's Law states that the flux through the circles is determined by the amount of charge inside. This can be rewritten as:

$$\oint \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\epsilon}$$

This is the integral form of Gauss's Law.

for empty space, $\epsilon = \epsilon_0$ (8.85e-12 farads per meter)

Wrap-Up

- Homework 4 released yesterday
- Next 2 studio sessions: Project 1